

Other trends like decline in LED prices, quick returns on investment (ROI) and technological innovations to keep prices low for a price-sensitive country like India are also driving this growth.

According to the Frost & Sullivan report, by 2018, demand for LED lighting products from Central and state government departments in India will be about 40 per cent of the total demand, and that from industries will be 60 per cent.

Explaining about how LED lighting market will grow in different verticals, Deepa Doraiswamy, industry manager, Electronics and Security, Frost & Sullivan, says, "By 2018, the highest growth in LED lighting market in India will be in the residential sector and railways at a CAGR of 82 per cent and 62 per cent, respectively, whereas the growth in the industrial sector and Central and state government departments will be at a CAGR of 48 per cent and 40 per cent, respectively. Streetlighting will grow at 42 per cent CAGR and commercial sector will move ahead at 39 per cent and automotive sector at 38 per cent CAGR."

Areas where demand is rising

Downlighters and streetlights. In India, government departments are the major consumers of LED lighting products, and the products that are most in demand are downlighters and streetlights. According to the Frost & Sullivan report, LED lighting is most beneficial for streetlighting and outdoor lighting applications because of a quicker ROI. According to the report, "The ROI of using LED streetlights has reduced from 6 to 7 years to 3 to 5 years, encouraging users to adopt them, and a large number of tenders have also been floated for the same in the ESCO mode."

"LED lighting technology is at a very nascent stage in India, but gaining momentum. The streetlights sector has seen a major movement in this space. The other sectors are commercial spaces, retail and hospitality. Currently, the conversion is driven by amount of usage (24-hour applications are most preferred). Today, in general lighting,

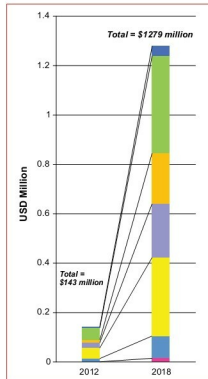
the percentage of home buyers for LEDs is still low due to high costs but this is changing rapidly and, according to a report by McKinsey, 70 per cent of lighting will become LED-based by 2020," says Arun Gupta.

Signage and display. The other area where demand is on the rise is signage and display. Besides metros like Delhi, Mumbai, Bengaluru and Chennai, the market in cities like Hyderabad, Ahmedabad, Jaipur and Pune have grown even faster and have registered a rise in demand. In Tier II and III cities, the continuous power shortages and load shedding are driving demand from the hospitality and emergency lighting sectors. LED displays and signage have a very high potential because of their multiple applications in different market segments, such as on airports, railway stations, trains, buses and metro trains. Not only this, demand for LED displays also comes from shopping malls and retail stores, clubs, multiplexes, trade fairs, hotels and hospitals, institutions, schools, banks, ATMs and for traffic information.

Government policies promoting LED lighting

Go-green initiatives. Recent government policies and initiatives are really giving a boost to the sector. Government played a very prominent role in creating awareness about energy efficiency and environmental friendliness of the LED lighting products. The Central and state governments' mandatory energy-efficiency regulations for new buildings and facilities, and incentive schemes for energy-efficient projects, have given the much-needed push to the LED lighting industry. "The go-green initiative, promoted by the government, will help LED penetrate the indoor and the outdoor applications of the government offices segment," says Deepa Doraiswamy.

As per an Electric Lamp and Component Manufacturers' Association of India (ELCOMA) report, the Ministry of Commerce has issued a notification to the Delhi government to ensure that all showrooms or display windows currently using incandescent or halo-



The growth forecast of the LED lighting market (2012-2018) (Source: Frost & Sullivan report)

gen lamps immediately replace them with LED downlighters.

The report also states that the Ministry of Commerce has also issued an advisory to the Ministry of Urban Development to ensure all medium and small municipalities change to LED streetlighting, and the Bureau of Energy Efficiency has started pilot programmes to encourage the use of LED streetlights for municipalities and local civic bodies for secondary roads, parks and parking areas. Not only this, government is also looking at the bulk requirement of railways, airport authority, urban housing, highways and other large development projects to create a large demand to encourage industry to manufacture. Government is also subsidising establishment of LED test facilities at the existing test labs. The subsidy could be up to ₹ 250 million.

Inclusion in PMA. In a major step, the government has also included LED products that utilise light produced by LEDs and cover applications like illumination, as well as optical displays like LED TVs, back lighting, signalling, indication and transportation,